STATISTICAL FOUNDATIONS FOR MACHINE LEARNING

Notes/Scribes for TA Session - Week 1

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An experiment consists of tossing a coin three times. What is the sample space of this experiment? Which event corresponds to the experiment resulting in more heads than tails?

Let **H** = **Heads** and **T** = **Tails**.

Sample space

When a coin is tossed 3 times, there are $2^3 = 8$ possible outcomes:

$$S = \{HHH,\ HHT,\ HTH,\ HTT,\ THH,\ THT,\ TTH,\ TTT\}$$

Event: more heads than tails

Since there are 3 tosses, "more heads than tails" means getting **2 or 3 heads**.

$$E = \{HHH,\ HHT,\ HTH,\ THH\}$$

Answer:

- Sample space: $\boxed{\{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}}$
- Event with more heads than tails: $\boxed{\{HHH, HHT, HTH, THH\}}$

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Let $S = \{1, 2, 3, 4, 5, 6, 7\}$, $E = \{1, 3, 5, 7\}$, $F = \{7, 4, 6\}$, $G = \{1, 4\}$. Find

(a) EF; **(c)** EG^c; **(e)** $E^c(F \cup G)$;

(b) $E \cup FG$; **(d)** $EF^c \cup G$; **(f)** $EG \cup FG$.

Given Sets and Complements

First, let's identify our base sets and their complements relative to S:

- $S = \{1, 2, 3, 4, 5, 6, 7\}$
- $E = \{1, 3, 5, 7\} \implies E^c = \{2, 4, 6\}$
- $F = \{4, 6, 7\} \implies F^c = \{1, 2, 3, 5\}$
- $G = \{1, 4\} \implies G^c = \{2, 3, 5, 6, 7\}$

Solutions

- **(a)** EF

  This is the intersection of E and F.

  $E \cap F = \{7\}$

- **(b)** $E \cup FG$

  First find $F \cap G = \{4\}$. Then union this with E.

  $E \cup \{4\} = \{1, 3, 4, 5, 7\}$

- **(c)** EG^c

  This is the intersection of E and G^c.

  $\{1, 3, 5, 7\} \cap \{2, 3, 5, 6, 7\} = \{3, 5, 7\}$

- **(d)** $EF^c \cup G$

  First find $E \cap F^c = \{1, 3, 5, 7\} \cap \{1, 2, 3, 5\} = \{1, 3, 5\}$. Then union this with G.

  $\{1, 3, 5\} \cup \{1, 4\} = \{1, 3, 4, 5\}$

- **(e)** $E^c(F \cup G)$

  First find $F \cup G = \{1, 4, 6, 7\}$. Then intersect this with E^c.

  $\{2, 4, 6\} \cap \{1, 4, 6, 7\} = \{4, 6\}$

- **(f)** $EG \cup FG$

  First find $E \cap G = \{1\}$ and $F \cap G = \{4\}$. Then take their union.

  $\{1\} \cup \{4\} = \{1, 4\}$

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Problem 4: Applications of probability properties

Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $A, B, C \in \mathcal{F}$. Suppose that

$$P(A) = 0.6, \quad P(B) = 0.5, \quad P(C) = 0.4,$$
$$P(A \cap B) = 0.3, \quad P(A \cap C) = 0.2, \quad P(B \cap C) = 0.2,$$

and

$$P(A \cap B \cap C) = 0.1.$$

Find the following probabilities:

(i)
$$P(A \cup B \cup C) \quad \text{and} \quad P(A^c \cap B^c \cap C^c).$$

(ii)
$$P\big((A \cup B) \cap C\big) \quad \text{and} \quad P\big(A \cup (B \cap C)\big).$$

(iii)
$$P\big((A^c \cup B^c) \cap C^c\big) \quad \text{and} \quad P\big((A^c \cap B^c) \cup C^c\big).$$

$$= 0.6 + 0.5 + 0.4 - 0.3 - 0.2 - 0.2 + 0.1$$

$$= \boxed{0.9}$$

2. Find $P(A^c \cap B^c \cap C^c)$

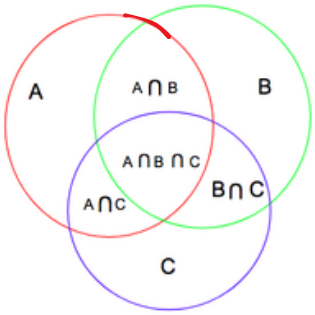
By De Morgan's law,

$$A^c \cap B^c \cap C^c = (A \cup B \cup C)^c$$

Therefore,

$$P(A^c \cap B^c \cap C^c) = 1 - P(A \cup B \cup C)$$

$$= 1 - 0.9 = \boxed{0.1}$$



$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

(ii)

1. Find $P((A \cup B) \cap C)$

Distribute the intersection:

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

Thus,

$$P((A \cup B) \cap C) = P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$$

$$= 0.2 + 0.2 - 0.1$$

$$= \boxed{0.3}$$

2. Find $P(A \cup (B \cap C))$

Using the addition rule:

$$P(A \cup (B \cap C)) = P(A) + P(B \cap C) - P(A \cap B \cap C)$$

$$= 0.6 + 0.2 - 0.1$$

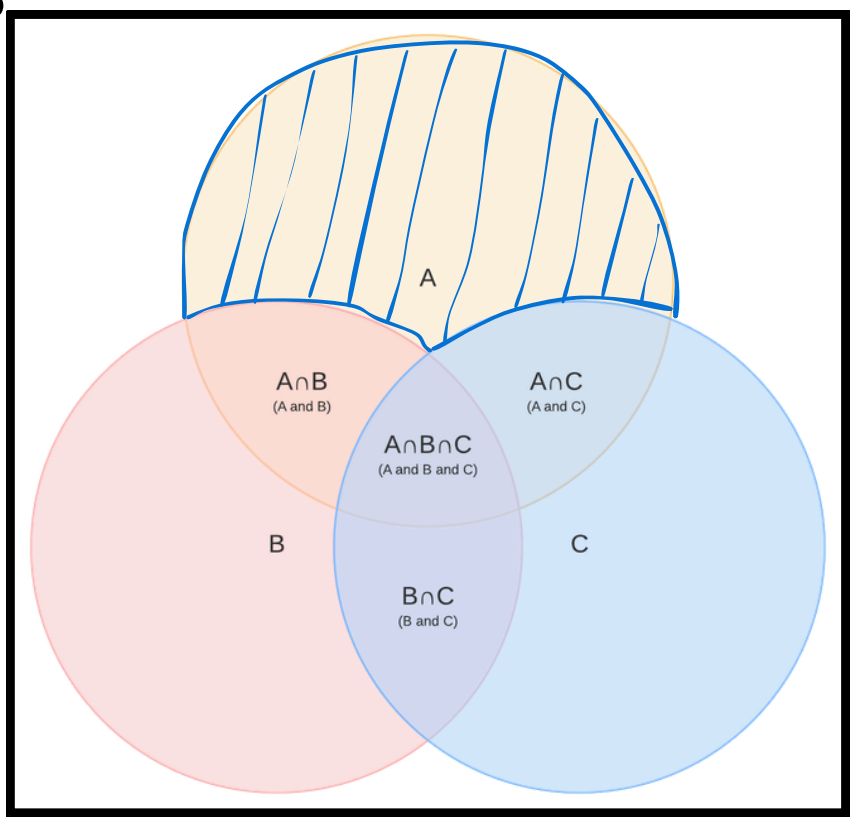
$$= \boxed{0.7}$$

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6. Let E, F, G be three events. Find expressions for the events that of E, F, G

(a) only E occurs;

S



and - intersection
or - union

(E Occurs, exclude F, G)

$$E \cap F^c \cap G^c.$$

To remove, exclude we'll be using intersections generally

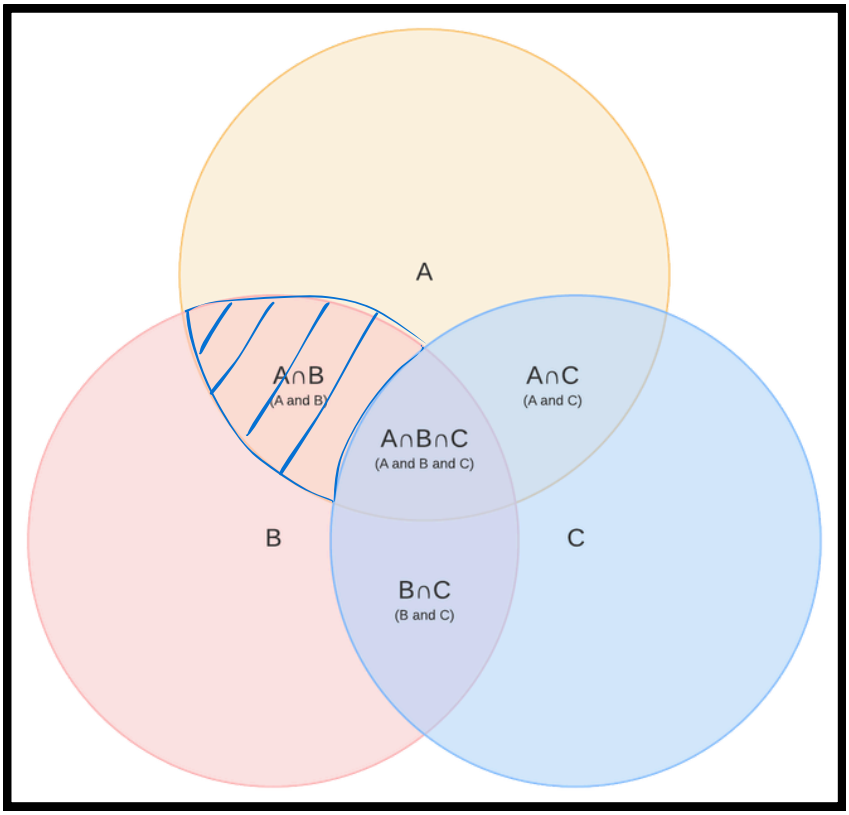
To remove a set,
just intersect with its complement

6. Let E, F, G be three events. Find expressions for the events that of E, F, G

(b) both E and G but not F occur;

E and G occur, remove F

$$E \cap G \cap F^c.$$



To remove a set,
just intersect with its complement

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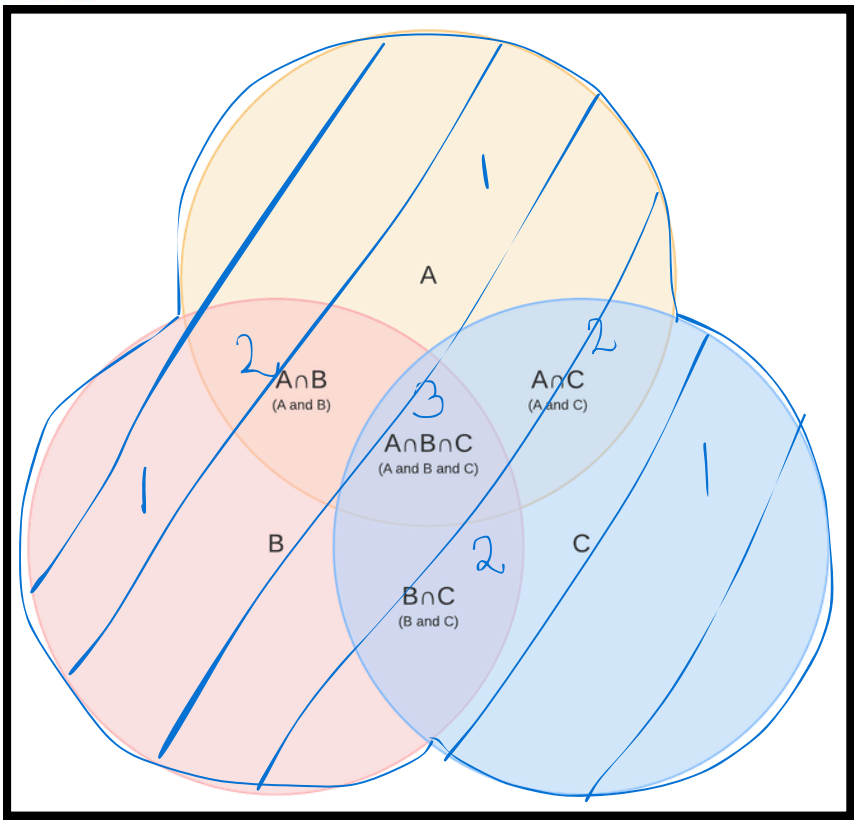
6. Let E, F, G be three events. Find expressions for the events that of E, F, G

(c) at least one of the events occurs;

At least = Minimum requirement
At most = Maximum requirement

E occurs or F occurs or G occurs

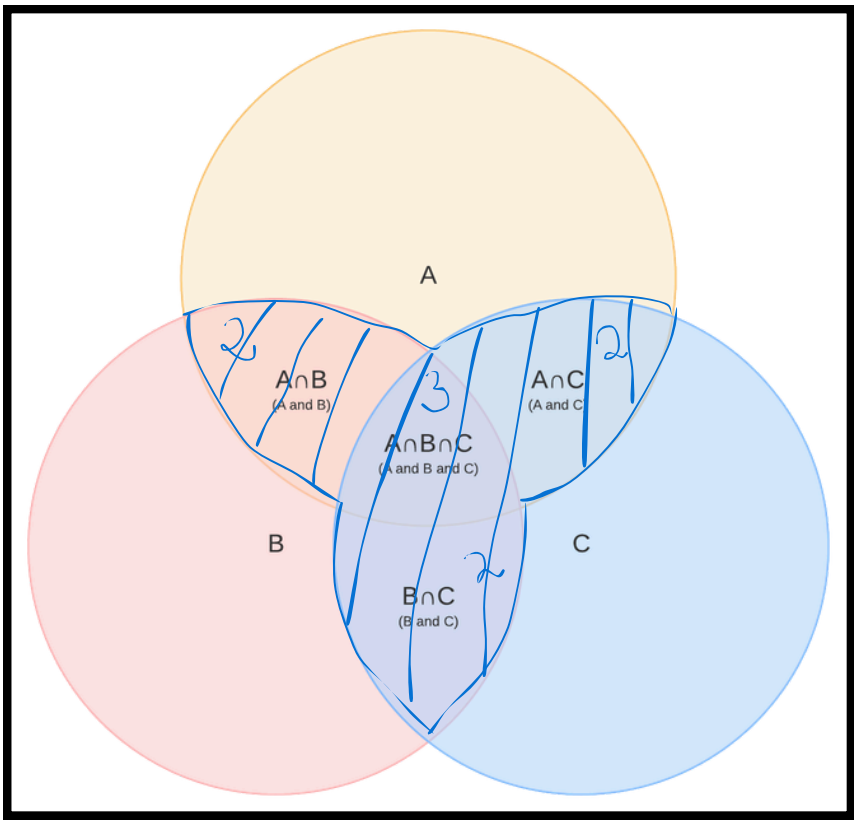
$$E \cup F \cup G.$$



(d) at least two of the events occur;

E and F occurs → $(E \cap F)$
or → $\cup$
F and G occurs → $(F \cap G)$
or → $\cup$
E and G occurs → $(E \cap G)$

$$(E \cap F) \cup (E \cap G) \cup (F \cap G).$$



whole flower

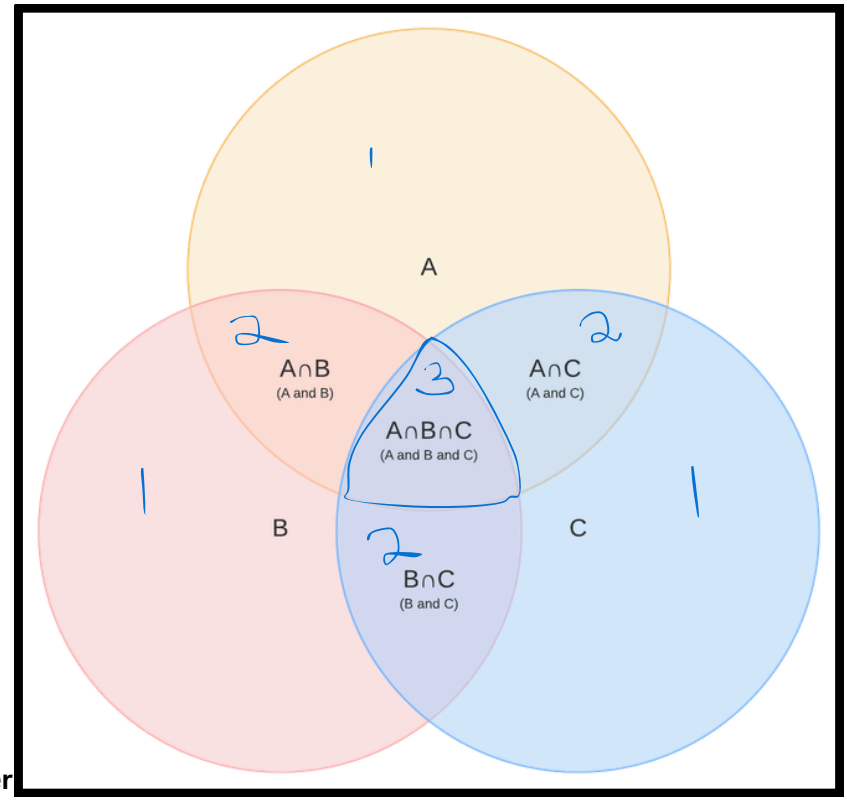
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6. Let E, F, G be three events. Find expressions for the events that of E, F, G

(e) all three occur;

E and F and G occurs

$$E \cap F \cap G.$$



middle of the flower

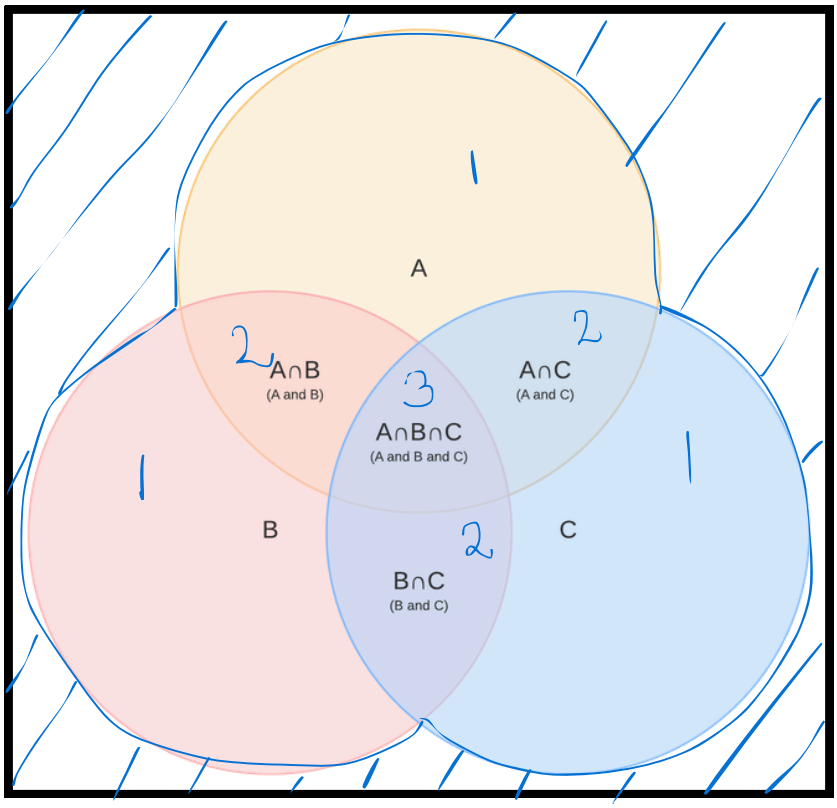
(f) none of the events occurs;

not E and not F and not G occurs

$$E^c \cap F^c \cap G^c$$

$$= (E \cup F \cup G)^c.$$

*both are correct and acceptable



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6. Let E, F, G be three events. Find expressions for the events that of E, F, G

(g) at most one of them occurs;

$$(E^c \cap F^c \cap G^c) \cup (E \cap F^c \cap G^c) \cup (E^c \cap F \cap G^c) \cup (E^c \cap F^c \cap G)$$

Nothing occurs or only E occurs or only F occurs or only G occurs

$$E^c \cap F^c \cap G^c \quad \cup \quad (E \cap F^c \cap G^c) \quad \cup \quad (E^c \cap F \cap G^c) \quad \cup \quad (E^c \cap F^c \cap G)$$

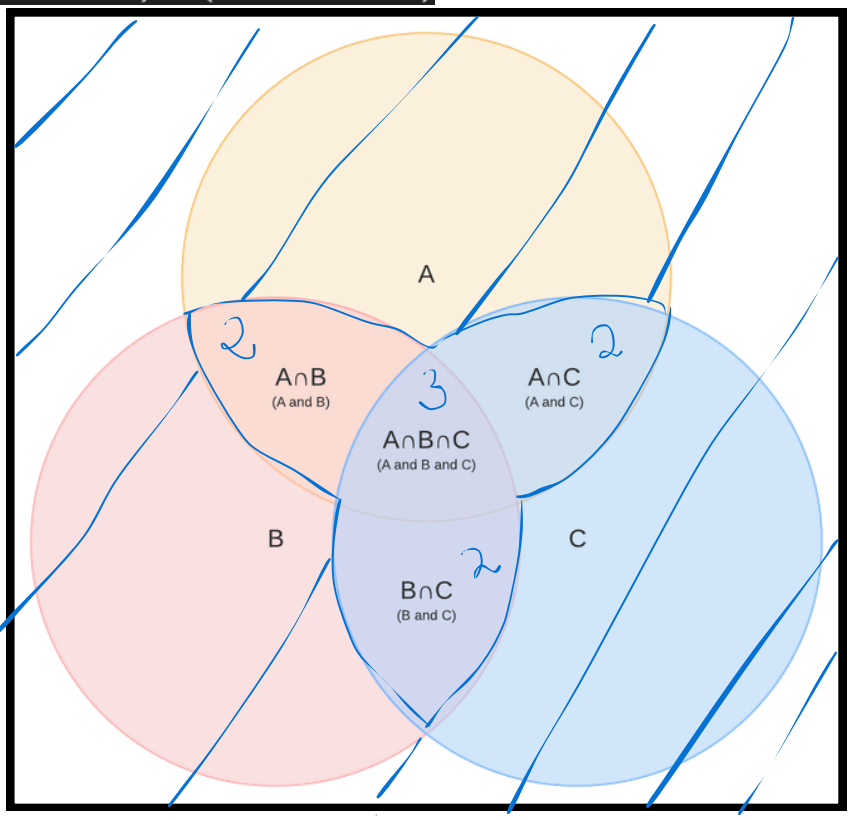
Whole flower was

$$(E \cap F) \cup (E \cap G) \cup (F \cap G).$$

Everything except the flower will be

$$((E \cap F) \cup (E \cap G) \cup (F \cap G))^c$$

*both are correct and acceptable



everything except the flower

(h) at most two of them occur;

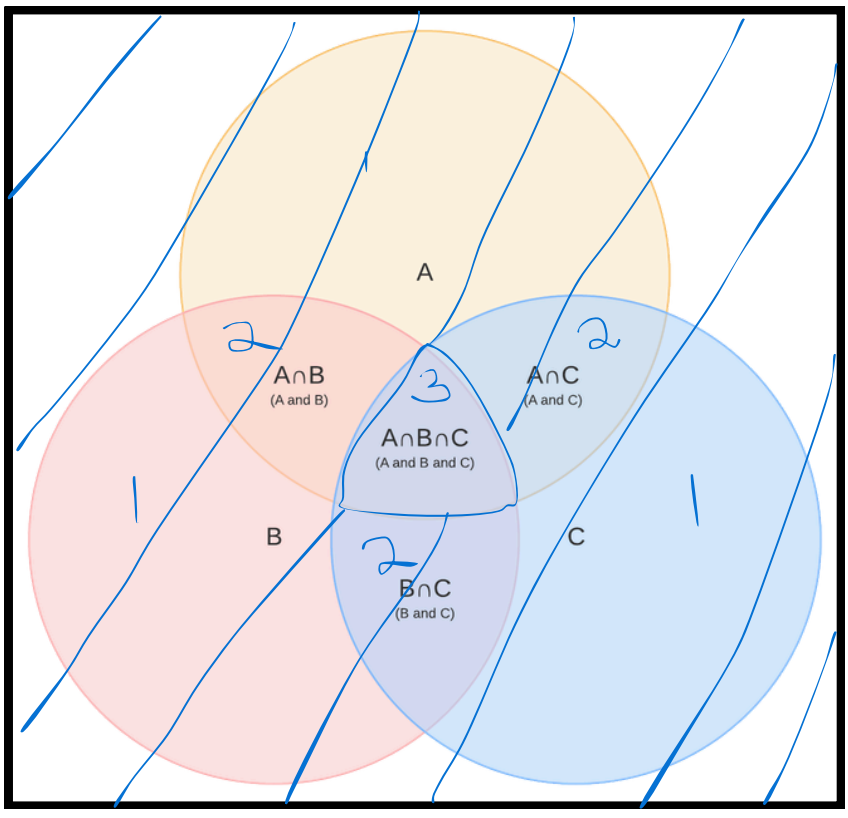
All three do not occur

Everything except "all three"

$$(E \cap F \cap G)^c$$

All three was

$$E \cap F \cap G$$



everything except the flower middle

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6. Let E, F, G be three events. Find expressions for the events that of E, F, G

(i) exactly two of them occur;

exactly E and F occurs $(E \cap F \cap G^c)$

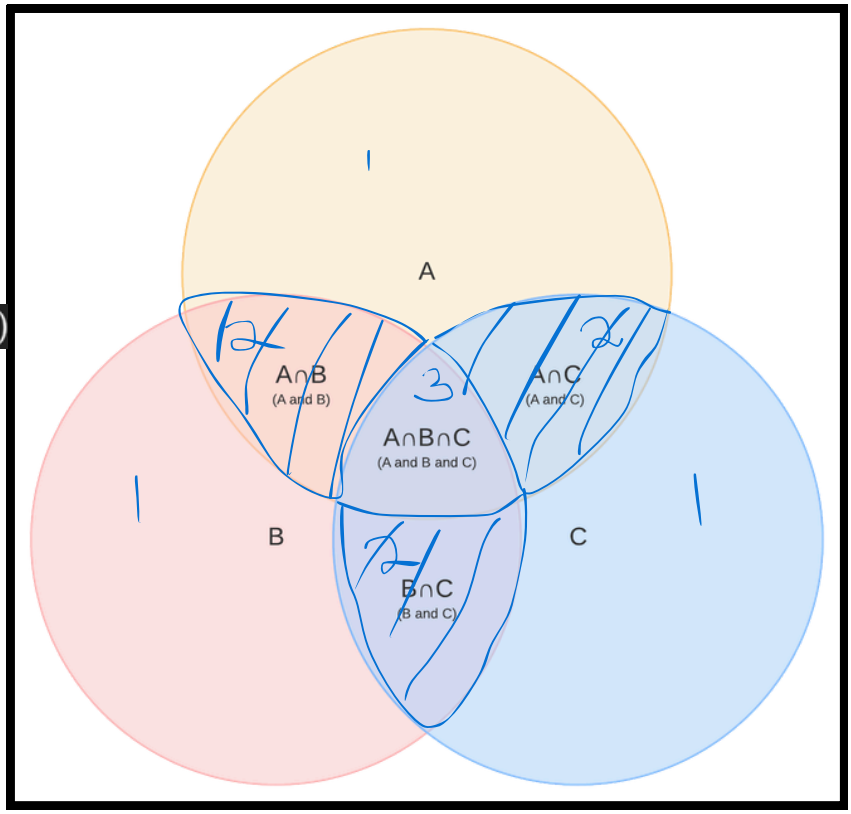
or

exactly F and G occurs $(F \cap G \cap E^c)$

or

exactly E and G occurs $(E \cap G \cap F^c)$

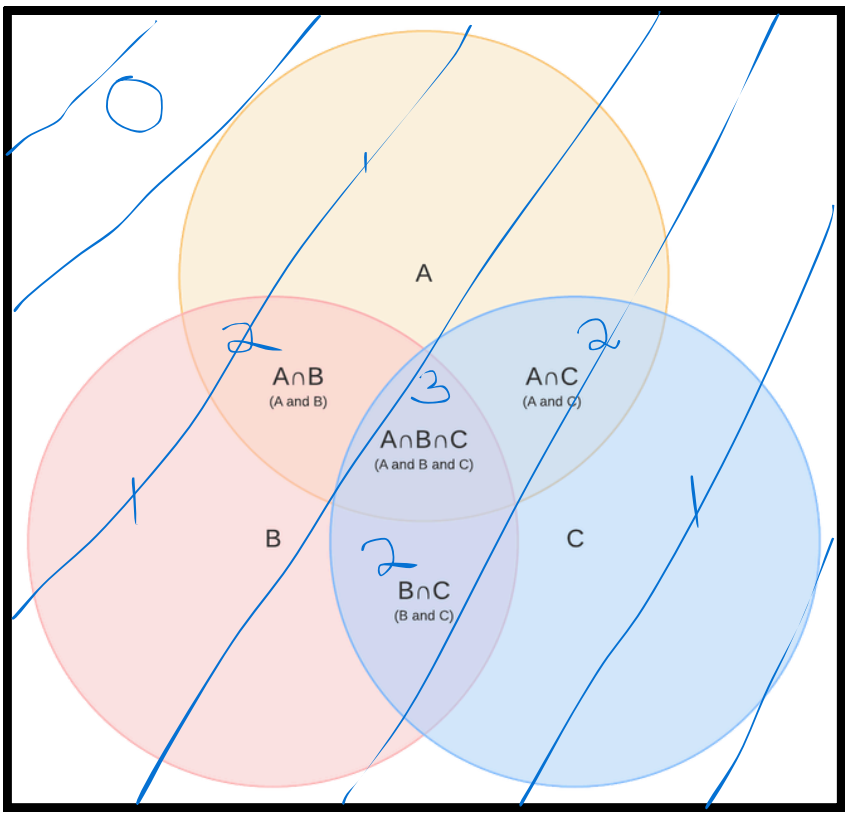
$$(E \cap F \cap G^c) \cup (E \cap G \cap F^c) \cup (F \cap G \cap E^c)$$



flower without the flower middle

(j) at most three of them occur.

Ω (the whole sample space)



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